

TABLES DI – 9-9-26

Directions for questions 1 to 5: These questions are based on the table given below.

The table shows the rates of interest as on 1st April of each year in three schemes X, Y and Z run by a finance company over 2008 to 2014.

Scheme → Year ↓	X	Y	Z
2008	16	18	19
2009	15	20	24
2010	18	24	30
2011	17	15	18
2012	17	20	10
2013	10	13	15
2014	12	14	16

Note: Given above rates of interest are being compounded annually. Also all the investments are made on 1st April of any year.

Solution:

Let the amount invested be ₹x.

$$x \times 1.1 \times 1.1 = 48400$$

$$\Rightarrow ₹40000.$$

Choice (C)

Amar invested Rs.20,000 in scheme X in 2010 for a year and subsequently invested the realized amount in scheme Y in 2011 for one year. The amount which he got from scheme Y was then re-invested in scheme Z in 2012 for a year. How much money did he get at the end of 2012?

☐ a) Rs.28954

☒ b) Rs.29854 (Correct Answer - ✓)

☐ c) Rs.29954

☐ d) Rs.24598

Solution:

$$\text{Required amount} = 20000 \times 1.18 \times 1.15 \times 1.1 \\ = 29854$$

Choice (B)

Sheena invested some amount in scheme Y in 2009. The amount she got after one year was re-invested in scheme Z in 2010 for a year. The amount that she finally got at the end of 2010, was Rs.31,200. How much did she invest in the beginning?

- ☐ a) Rs.20,000 (Correct Answer - ✓)
- ☐ b) Rs.22,000
- ☐ c) Rs.24,000
- ☐ d) Rs.25,000

Solution:

$$\text{Let the amount invested be ₹}x. \\ x \times 1.2 \times 1.3 = 31200 \\ \Rightarrow x = 20000.$$

Choice (A)

Dheeraj invested some amount in the ratio 3 : 4 in the schemes X and Y respectively in 2008 and subsequently invested the amount he got after one year in scheme Z for 2 years. The final amount was Rs.82,615. What is the money he had in the beginning?

- ☐ a) Rs.40,000
- ☐ b) Rs.42,000
- ☐ c) Rs.42,350

- ☐ d) Rs.43,750 (Correct Answer - ✓)

Solution:

$$\begin{aligned} &\text{Let the amount be } 3x \text{ and } 4x. \\ &(3x \times 1.16 + 4x \times 1.18)(1.24)(1.3) \\ &\frac{41}{5}x \times \frac{961}{625} = 82,615 \\ &x = ₹6250 \\ &\text{Money he had} = 7x \\ &= 43750. \end{aligned}$$

Choice (D)

What is the difference between the interest earned on a sum of Rs.20,000 if it is invested in scheme X in 2011 for 2 years and in scheme Y for the same period?

☐ a) Rs.111

☐ b) Rs.333

☐ c) Rs.444

☒ d) Rs.222 (Correct Answer - ✓)

Solution:

$$\begin{aligned} \text{Difference} &= 20000 (1.15 \times 1.2 - 1.17^2) \\ &= 222. \end{aligned}$$

Choice (D)

Directions for questions 6 to 10: These questions are based on the table given below.

The table gives data pertaining to the number of male and female employees who joined and left ABC corp. In 2014, at all their branches in different locations.

	2014 (Joined)		2014 (Left)	
	Males	Females	Males	Females
Kochi	119	84	32	38
Hyderabad	246	172	64	50
Delhi	430	310	104	76
Bangalore	600	64	170	150
Shimla	186	224	44	30
Dubai	560	440	96	72
U.A.E	640	600	88	80

In 2014, the number of females who left the Dubai branch is what percentage of the females who joined the same branch?

☐ a) $15\frac{4}{11}\%$

☒ b) $16\frac{4}{11}\%$ (Correct Answer - ✓)

☐ c) $17\frac{3}{11}\%$

☐ d) $18\frac{4}{11}\%$

Solution:

$$\begin{aligned}\text{Required percentage} &= \frac{72}{440} \times 100 \\ &= 16\frac{4}{11}\% . \quad \text{Choice (B)}\end{aligned}$$

The total number of females who left from all the branches is approximately what percentage of the total number of females who joined in all the branches?

☒ a) 26% (Correct Answer - ✓)

☐ b) 30%

☐ c) 35%

☐ d) 40%

Solution:

$$\text{Required percentage} = \frac{38+50+76+150+30+72+80}{84+172+310+64} = \frac{248}{947} \cong 26\%.$$

Choice (A)

What is the ratio of the least number of males who joined any branch to the highest number of males who left any branch?

- ☐ a) 7 : 10
- ☐ b) 10 : 7
- ☐ c) 7 : 11
- ☐ d) 11 : 7

Solution:

$$\text{Ratio} = \frac{119}{170} = \frac{7}{10}.$$

Choice (A)

What is the net change in the number of males working in all the branches of ABC corp in 2014?

- ☐ a) 2172
- ☐ b) 2203
- ☐ c) 2322

☐ d) 2183 (Correct Answer - ✓)

Solution:

$$\begin{aligned} &\text{Resultant number of males} \\ &= \text{Number of males joined} - \text{Number of males left} \\ &= (119 - 32) + (246 - 64) + (430 - 104) + \\ &\quad (600 - 170) \\ &\quad + (186 - 44) + (560 - 96) + (640 - 88) \\ &= 2183. \end{aligned}$$

Choice (D)

The total number of females who left from all the branches is approximately what percentage of the total number of males who joined all the branches?

☐ a) 22%

☐ b) 25%

☒ c) 18% (Correct Answer - ✓)

☐ d) 30%

Solution:

$$\text{Required percentage} = \frac{38+50+76+150+30+72+80}{119+246+430+600+186+560+640} \cong 18\% .$$

Choice (C)

Directions for questions 11 to 15: These questions are based on the information given below.

Vinod Melkote, HR Manager of XYZ Ltd, received partial information from his subordinates about the employees for the annual appraisal.

	Performance			Total
	Below Average	Average	Above Average	
Males				96
Females		30		
Total	48			

Vinod also received the following information.

- A. One – third of the total number of employees are above – average performers.
- B. 25% of the males are below – average performers.
- C. The total number of females is twice the number of average performing males.

Directions for questions 11 to 14: Type in your answer in the input box provided below the question.

How many people are above- average performers?

Solution:

Given that 25% of the males are below – average performers. Hence their number is 25% of 96 = 24.

As the total number of below – average performers is 48 and 24 of them are males, the remaining 24 are females.

Given, the number of females = 2 (Number of male average performers)

Let the number of male average performers be x. Then, we get the following table.

	Below Average	Average	Above Average	Total
Males	24	x	72 – x	96
Females	24	30	2x – 54	2x
Total	48	30 + x	x + 18	2x + 96

Given, above average performers = $\frac{1}{3}$

(total number of employees)

$$\text{i.e., } x + 18 = \frac{1}{3} (2x + 96)$$

$$\Rightarrow 3x + 54 = 2x + 96$$

$$\Rightarrow x = 42$$

The number of above – average performers = x + 18 i.e. 60.

Ans : (60)

Directions for questions 11 to 14: Type in your answer in the input box provided below the question.

How many females are there?

Solution:

Given that 25% of the males are below – average performers. Hence their number is 25% of 96 = 24.

As the total number of below – average performers is 48 and 24 of them are males, the remaining 24 are females.

Given, the number of females = 2 (Number of male average performers)

Let the number of male average performers be x . Then, we get the following table.

	Below Average	Average	Above Average	Total
Males	24	x	$72 - x$	96
Females	24	30	$2x - 54$	$2x$
Total	48	$30 + x$	$x + 18$	$2x + 96$

Given, above average performers = $\frac{1}{3}$

(total number of employees)

$$\text{i.e., } x + 18 = \frac{1}{3}(2x + 96)$$

$$\Rightarrow 3x + 54 = 2x + 96$$

$$\Rightarrow x = 42$$

The number of females = $2x$

$$\text{i.e. } 2(42) = 84$$

Ans : (84)

Directions for questions 11 to 14: Type in your answer in the input box provided below the question.

How many females can be classified as above – average performers?

Solution:

Given that 25% of the males are below – average performers. Hence their number is 25% of 96 = 24.

As the total number of below – average performers is 48 and 24 of them are males, the remaining 24 are females.

Given, the number of females = 2 (Number of male average performers)

Let the number of male average performers be x . Then, we get the following table.

	Below Average	Average	Above Average	Total
Males	24	x	$72 - x$	96
Females	24	30	$2x - 54$	$2x$
Total	48	$30 + x$	$x + 18$	$2x + 96$

Given, above average performers = $\frac{1}{3}$

(total number of employees)

$$\text{i.e., } x + 18 = \frac{1}{3}(2x + 96)$$

$$\Rightarrow 3x + 54 = 2x + 96$$

$$\Rightarrow x = 42$$

The number of females who are above average performers = $2x - 54$ i.e. $2(42) - 54$
= 30
Ans : (30)

Directions for questions 11 to 14: Type in your answer in the input box provided below the question.

What is the difference between the total number of average performers and males?

Solution:

Given that 25% of the males are below – average performers. Hence their number is 25% of 96 = 24.

As the total number of below – average performers is 48 and 24 of them are males, the remaining 24 are females.

Given, the number of females = 2 (Number of male average performers)

Let the number of male average performers be x . Then, we get the following table.

	Below Average	Average	Above Average	Total
Males	24	x	$72 - x$	96
Females	24	30	$2x - 54$	$2x$
Total	48	$30 + x$	$x + 18$	$2x + 96$

Given, above average performers = $\frac{1}{3}$

(total number of employees)

$$\text{i.e., } x + 18 = \frac{1}{3} (2x + 96)$$

$$\Rightarrow 3x + 54 = 2x + 96$$

$$\Rightarrow x = 42$$

The total number of average performers = $x + 30$ i.e. 72

Also as the number of males = 96, the required difference is $96 - 72 = 24$

Ans : (24)

Directions for question 15: Select the correct alternative from the given choices.

What is the ratio of the number of average performers to the number of females?

☒ a) 6 : 7 (Correct Answer - ✓)

☐ b) 11 : 12

☐ c) 3 : 4

☐ d) 2 : 3

Solution:

Given that 25% of the males are below – average performers. Hence their number is 25% of 96 = 24.

As the total number of below – average performers is 48 and 24 of them are males, the remaining 24 are females.

Given, the number of females = 2 (Number of male average performers)

Let the number of male average performers be x . Then, we get the following table.

	Below Average	Average	Above Average	Total
Males	24	x	$72 - x$	96
Females	24	30	$2x - 54$	$2x$
Total	48	$30 + x$	$x + 18$	$2x + 96$

Given, above average performers = $\frac{1}{3}$

(total number of employees)

$$\text{i.e., } x + 18 = \frac{1}{3} (2x + 96)$$

$$\Rightarrow 3x + 54 = 2x + 96$$

$$\Rightarrow x = 42$$

The required ratio of above average performers:

The number of females = $x + 30 : 2x$ i.e. $72 : 84 = 6 : 7$

Choice (A)